

10.1 NATURE OF LIGHT : INTRODUCTORY CONCEPTS

1. *Briefly discuss the various theories about the nature of light.*

Nature of light. Various theories about the nature of light have been proposed from time to time. Some of the main theories are as follows :

1. **Corpuscular theory of light.** *Newton*, the great among the greatest, proposed in 1675 A.D. that light consists of tiny particles called *corpuscles* which are shot out at high speed by a luminous object. This theory could explain the reflection, refraction and rectilinear propagation of light.

2. **Wave theory of light.** In 1678, Dutch scientist *Christian Huygens*, suggested that light travels in the form of longitudinal waves just as sound propagates through air. He proposed that light waves propagate through an all-pervading hypothetical medium, called *luminiferous ether*. Later on, the existence of such a medium was discarded due to its contradictory properties. *Fresnel* and *Young* showed that light propagates as a transverse wave. This successfully explained the reflection, refraction as well as interference, diffraction and polarisation of light waves.

3. **Electromagnetic nature of light waves.** In 1873, *Maxwell* suggested that light propagates as electric

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and magnetic field oscillations. These are called electromagnetic waves which require no medium for their propagation. Also, these waves are transverse in nature.

4. *Planck's quantum theory of light.* According to *Max Planck*, light travels in the form of small packets of energy called photons. In 1905, *Albert Einstein* used this theory to explain photoelectric effect (emission of electrons from a metal surface when light falls on it).

So we see that in phenomena like interference, diffraction and polarisation, light behaves as a wave while in photoelectric effect, it behaves as a particle. *de Broglie* suggested that *light has a dual nature, i.e., it can behave as particles as well as waves.*

10.2 WAVEFRONTS AND RAYS

2. *Distinguish between a wavefront and a ray of light. What are spherical, cylindrical and plane wavefronts ? Give their examples, sketch wavefronts corresponding to parallel, converging and diverging rays of light.*

Wavefronts. Suppose a stone is thrown on the surface of still water. Circular patterns of alternate crests and troughs begin to spread out from the point of impact. Clearly, all the particles lying on a crest are in the position of their maximum upward displacement and hence in the same phase. Similarly, all particles lying on a trough are in the position of their maximum downward displacement and, therefore, in

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the same phase. The locus of all such points oscillating in the same phase is called a **wavefront**. Thus every crest or a trough is a wavefront.

A wavefront is defined as the continuous locus of all such particles of the medium which are vibrating in the same phase at any instant.

Thus a wavefront is a surface of constant phase. The speed with which the wavefront moves outwards from the source is called the **phase speed**.

Different types of wavefronts. The geometrical shape of a wavefront depends on the source of disturbance. Some of the common shapes are :

1. Spherical wavefront. In the case of waves travelling in all directions from a point source, the wavefronts are spherical in shape. This is because all such points which are equidistant from the point source will lie on a sphere [Fig. 10.1(a)] and the disturbance starting from the source-S will reach all these points simultaneously.

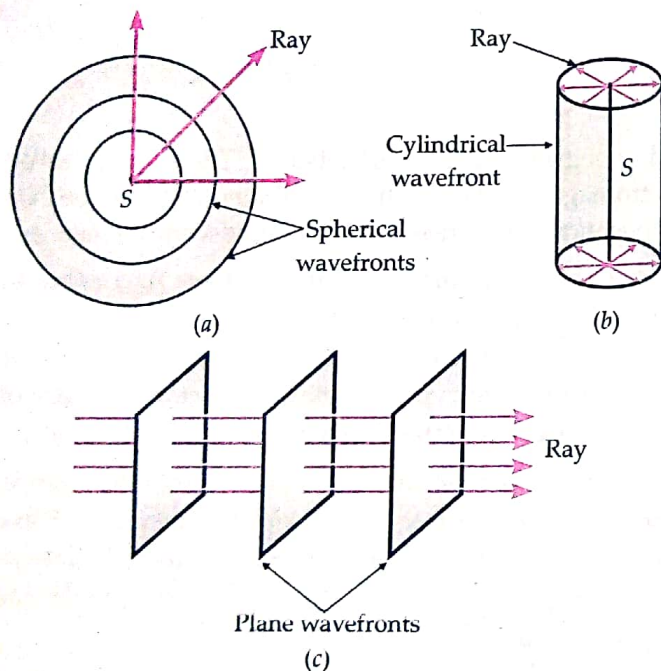


Fig. 10.1 Different types of wavefronts.

2. Cylindrical wavefront. When the source of light is linear in shape, such as a fine rectangular slit, the wavefront is cylindrical in shape. This is because the locus of all such points which are equidistant from the linear source will be a cylinder [Fig. 10.1(b)].

3. Plane wavefront. As a spherical or cylindrical wavefront advances, its curvature decreases progressively. So a small portion of such a wavefront at a large distance from the source will be a plane wavefront [Fig. 10.1(c)].

Ray of light. It is seen that whatever is the shape of wavefront, the disturbance travels outwards along

straight lines emerging from the source, i.e., the energy of a wave travels in a direction perpendicular to the wavefront.

An arrow drawn perpendicular to a wavefront in the direction of propagation of a wave is called a **ray**.

A ray of light represents the path along which light travels.

If we measure the separation between a pair of wavefronts along any ray, it is found to be a constant.

This illustrates two general principles :

1. Rays are perpendicular to wavefronts.
2. The time taken for light to travel from one wavefront to another is the same along any ray.

In case of a plane wavefront, the rays are parallel [Fig. 10.2(a)]. A group of parallel rays is called a **beam of light**. In case of a spherical wavefront, the rays either converge to a point [Fig. 10.2(b)] or diverge from a point [Fig. 10.2(c)].

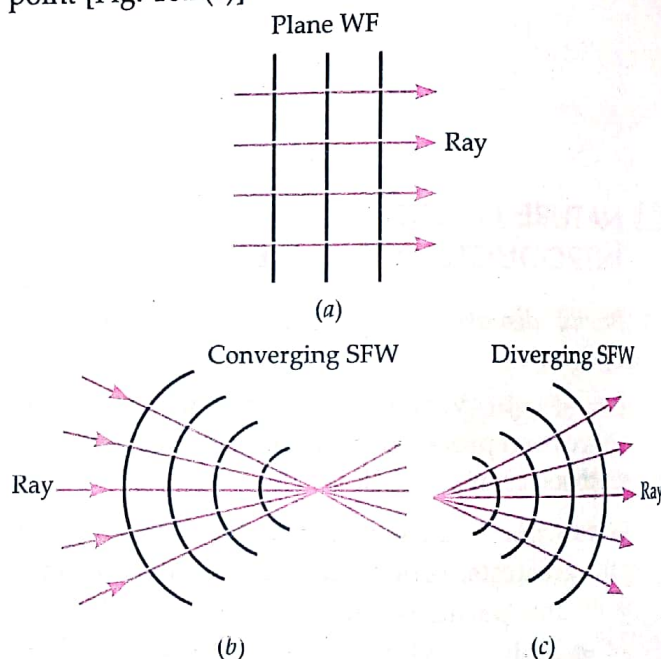


Fig. 10.2 Wavefronts and corresponding rays in three cases (a) plane, (b) converging spherical and (c) diverging spherical.

10.3 HUYGENS' PRINCIPLE OF SECONDARY WAVELETS

3. State the assumptions on which Huygens' principle of secondary wavelets is based. Describe Huygens' construction for the propagation of wavefronts in a medium. Is a backward wavefront also possible ?

Huygens' principle. Huygens' principle is the basis of wave theory of light. It tells how a wavefront propagates through a medium. According to Huygens' principle, each point on a wavefront is a source of secondary waves, which add up to give a wavefront at any later time.

This principle is based on the following assumptions:

1. Each point on a wavefront acts as a fresh source of new disturbance, called secondary waves or wavelets.
2. The secondary wavelets spread out in all directions with the speed of light in the given medium.
3. The new wavefront at any later time is given by the forward envelope (tangential surface in the forward direction) of the secondary wavelets at that time.

Huygens' construction. It is a geometrical method of locating the new position and shape of a wavefront at any instant from its known position and shape at any other instant. The various steps involved are as follows:

1. Consider a spherical [Fig. 10.3(a)] or plane [Fig. 10.3(b)] wavefront moving towards right. Let AB be its position at any instant of time. The region on its left has received the wave while region on the right is undisturbed.

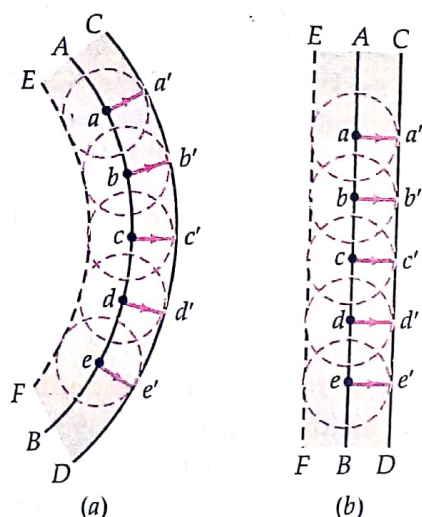


Fig. 10.3 Huygens' geometrical construction for the propagation of (a) spherical, (b) plane wavefront.

2. According to Huygens' principle, each point on AB becomes a source of secondary disturbance, which travels with the same speed c . To find the new wavefront after time t , we draw spheres of radii ct , from each point on AB.

3. The forward envelope or the tangential surface CD of the secondary wavelets gives the new wavefront after time t .

4. The lines aa' , bb' , cc' , etc., are perpendicular to both AB and CD. Along these lines, the energy flows from AB to CD. So these lines represent the rays. Rays are always normal to wavefronts.

No backward wavefront is possible. There cannot be backward flow of energy during the propagation of a wave. It can be shown mathematically that the amplitude of secondary wavelets is proportional to

$(1 + \cos \theta)$, where θ is the angle between the ray at the point of consideration and the direction of secondary wavelets. For a backward wavefront $\theta = \pi$, so that $1 + \cos \theta = 0$. Thus the resultant amplitude of all the secondary wavelets at any point on the backward wavefront is zero. In fact, the effects of secondary wavelets cancel out at all points except those lying on the forward envelope. So a backward wavefront cannot exist.

Huygens' principle of secondary wavelets can be used to prove the laws of reflection and refraction.

For Your Knowledge

- A wavefront is a surface of constant phase.
- A ray of light is the path along which light travels. It is always normal to the wavefront.
- In a homogeneous and isotropic medium (i.e., a medium having uniform composition and the same properties in all directions), the speed of light is same in all directions and the secondary wavelets are spherical. The rays are then perpendicular to both the wavefronts and the time of travel measured along any ray from one wavefront to the next is always same.
- Only the front portions of the secondary wavelets add up to give rise to a wavefront in the forward direction. The back portions of the secondary wavelets add up to zero, so no backward wavefront is possible.

10.4 REFLECTION ON THE BASIS OF WAVE THEORY

4. Derive the laws of reflection of light on the basis of Huygens' wave theory of light.

Laws of reflection on the basis of Huygens' wave theory. As shown in Fig. 10.4, consider a plane wavefront AB incident on the plane reflecting surface XY, both the wavefront and the reflecting surface being perpendicular to the plane of paper.

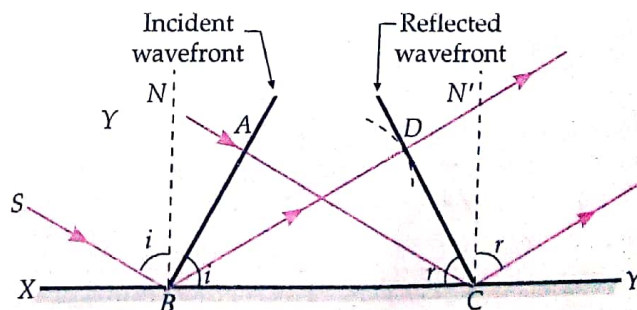


Fig. 10.4 Wavefronts and corresponding rays for reflection from a plane surface.

First the wavefront touches the reflecting surface at B and then at the successive points towards C. In accordance with Huygens' principle, from each point on BC, secondary wavelets start growing with the

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speed c . During the time the disturbance from A reaches the point C , the secondary wavelets from B must have spread over a hemisphere of radius $BD = AC = ct$, where t is the time taken by the disturbance to travel from A to C . The tangent plane CD drawn from the point C over this hemisphere of radius ct will be the *new reflected wavefront*.

Let angles of incidence and reflection be i and r respectively. In $\triangle ABC$ and $\triangle DCB$, we have

$$\angle BAC = \angle CDB \quad [\text{Each is } 90^\circ]$$

$$BC = BC \quad [\text{Common}]$$

$$AC = BD \quad [\text{Each is equal to } ct]$$

$$\therefore \triangle ABC \cong \triangle DCB$$

$$\text{Hence } \angle ABC = \angle DCB$$

$$\text{or } \angle i = \angle r$$

i.e., the angle of incidence is equal to the angle of reflection.
This proves the first law of reflection.

Further, since the incident ray SB , the normal BN and the reflected ray BD are respectively perpendicular to the incident wavefront AB , the reflecting surface XY and the reflected wavefront CD (all of which are perpendicular to the plane of the paper), therefore, they all lie in the plane of the paper, *i.e.*, in the same plane. This proves the second law of reflection.

10.5 REFRACTION ON THE BASIS OF WAVE THEORY

5. Derive the laws of refraction of light on the basis of Huygens' wave theory of light.

Laws of refraction on the basis of Huygens' wave theory. Consider a plane wavefront AB incident on a plane surface XY , separating two media 1 and 2, as shown in Fig. 10.5. Let v_1 and v_2 be the velocities of light in the two media, with $v_2 < v_1$.

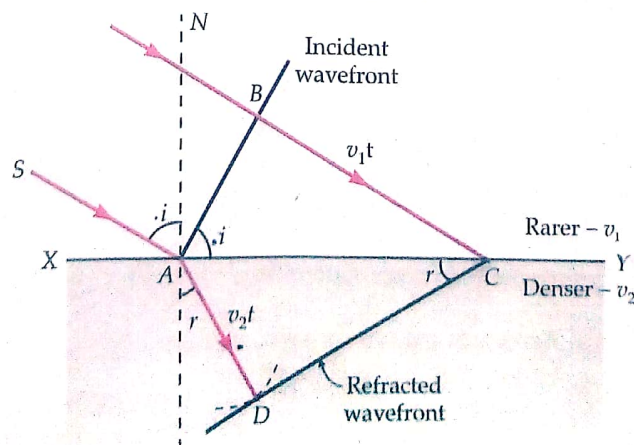


Fig. 10.5 Wavefronts and corresponding rays for refraction by a plane surface separating two media.

The wavefront first strikes at point A and then at the successive points towards C. According to Huygens' principle, from each point on AC, the secondary wavelets start growing in the second medium with speed v_2 . Let the disturbance take time t to travel from B to C, then $BC = v_1 t$. During the time the disturbance from B reaches the point C, the secondary wavelets from point A must have spread over a hemisphere of radius $AD = v_2 t$ in the second medium. The tangent plane CD drawn from point C over this hemisphere of radius $v_2 t$ will be the new refracted wavefront.

Let the angles of incidence and refraction be i and r respectively.

From right $\triangle ABC$, we have

$$\sin \angle BAC = \sin i = \frac{BC}{AC}$$

From right $\triangle ADC$, we have

$$\sin \angle DCA = \sin r = \frac{AD}{AC}$$

$$\therefore \frac{\sin i}{\sin r} = \frac{BC}{AD} = \frac{v_1 t}{v_2 t}$$

$$\text{or} \quad \frac{\sin i}{\sin r} = \frac{v_1}{v_2} = {}^1\mu_2 \text{ (a constant)}$$

This proves *Snell's law of refraction*. The constant ${}^1\mu_2$ is called the *refractive index* of the second medium with respect to first medium.

Further, since the incident ray SA, the normal AN and the refracted ray AD are respectively perpendicular to the incident wavefront AB, the dividing surface XY and the refracted wavefront CD (all perpendicular to the plane of the paper), therefore, they all lie in the plane of the paper, i.e., in the same plane. This proves another law of refraction.

10.8 PRINCIPLE OF SUPERPOSITION OF WAVES

8. State and explain the principle of superposition of waves.

Principle of superposition of waves. When a number of waves travel through a medium simultaneously, each wave travels independently of the others i.e., as if all other waves were absent. An important consequence of this independent behaviour of the waves is that the effects of all these waves get added together. The resultant wave is obtained by the **principle of superposition of waves** which can be stated as follows :

When a number of waves travelling through a medium superpose on each other, the resultant displacement at any point at a given instant is equal to the vector sum of the displacements due to the individual waves at that point.

If $\vec{y}_1, \vec{y}_2, \vec{y}_3, \dots, \vec{y}_n$ are the displacements due to the different waves acting separately, then according to the principle of superposition, the resultant displacement when all the waves act together is given by the vector sum :

$$\vec{y} = \vec{y}_1 + \vec{y}_2 + \vec{y}_3 + \dots + \vec{y}_n$$

When the two superposing waves are in the same phase i.e., the crest of one falls over the crest of another [Fig. 10.10(a)] or the trough of one falls over the trough of another [Fig. 10.10(b)], their displacements get added. When the two waves meet in opposite phases i.e., the crest of one falls over the trough of another [Fig. 10.10(c)], their displacements get subtracted.

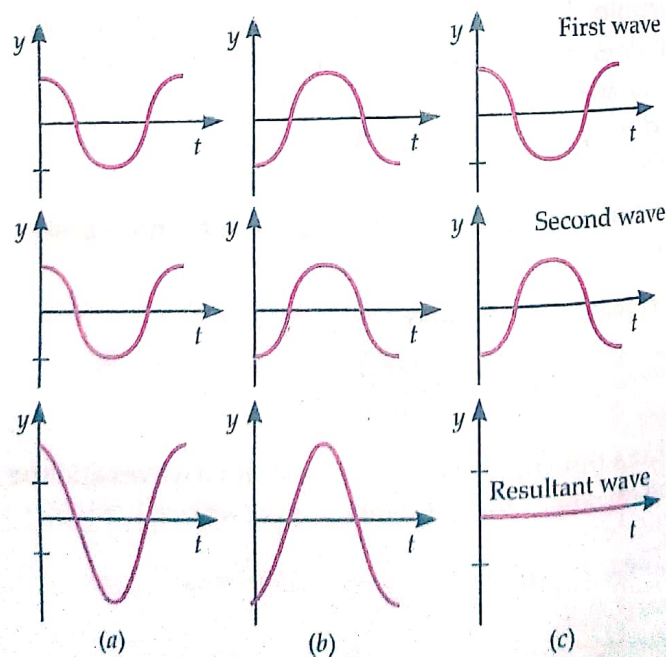


Fig. 10.10 Illustration of principle of superposition of waves.

10.9 INTERFERENCE OF LIGHT : YOUNG'S DOUBLE SLIT EXPERIMENT

9. What is interference of light? Describe Young's double slit experiment for observing the interference of light.

Interference of light. When two light waves of the same frequency and having zero or constant phase difference travelling in the same direction superpose each other, the intensity in the region of superposition gets redistributed, becoming maximum at some points and minimum at others. This phenomenon is called **interference of light**.

Young's double slit experiment. In 1801, Thomas Young was the first person to demonstrate experimentally the interference of light.

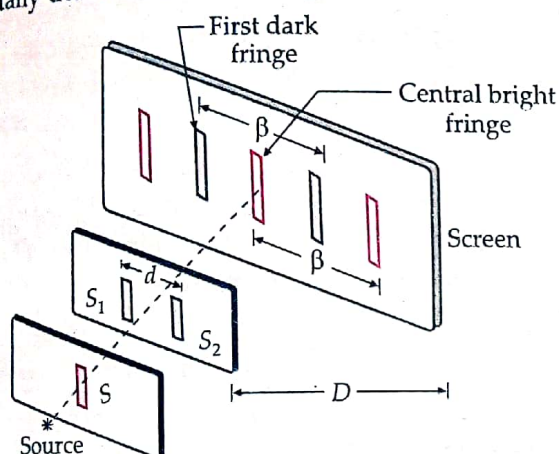


Fig. 10.11 Young's double slit experiment.

In this experiment, a source of monochromatic light (e.g., a sodium vapour lamp) illuminates a rectangular narrow slit S , about 1 mm wide, as shown in Fig. 10.11. S_1 and S_2 are two parallel narrow slits which are arranged symmetrically and parallel to the slit S at a distance of about 10 cm from it. The separation between S_1 and S_2 is ≈ 2 mm and width of each slit is ≈ 0.3 mm. An observation screen is placed at a distance of ≈ 2 m from the two slits. Alternate bright and dark bands appear on the observation screen. These are called **interference fringes**. When one of the slits, S_1 or S_2 is closed, bright and dark fringes disappear and the intensity of light becomes uniform.

Explanation. Fig. 10.12 shows a section of Young's experiment in the plane of paper. According to Huygens' principle, cylindrical wavefronts emerge out from slit S , whose sections have been shown by circular arcs. The solid curves represent crests and the dotted curves represent troughs. As $SS_1 = SS_2$, these waves fall on the slits S_1 and S_2 simultaneously so that the waves spreading out from S_1 and S_2 are in the same phase. Thus S_1 and S_2 act as two coherent sources of monochromatic light. Interference takes place between the waves diverging from these sources.

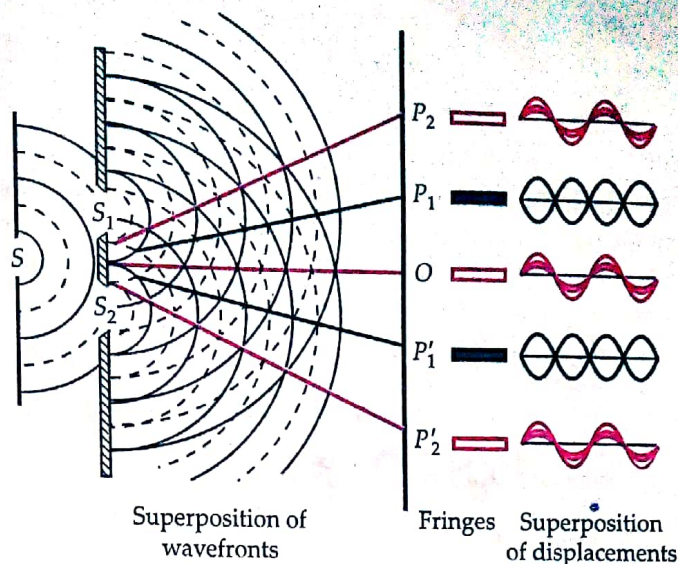


Fig. 10.12 Interference of two light beams.

At the lines leading to O , P_2 and P'_2 , the crest of one wave falls over the crest of other wave or the trough of one wave falls over the trough of other wave, the amplitudes of the two waves get added up and hence the intensity ($I \propto a^2$) becomes maximum. This is called **constructive interference**. At the lines leading to P_1 and P'_1 , the crest of one wave falls over the trough of other or the trough of one wave falls over the crest of other wave, the amplitudes of the two waves get subtracted and hence the intensity becomes minimum. This is called **destructive interference**. So on the observation screen, we obtain a number of alternate bright and dark fringes, parallel to the two slits.

10.10 CONDITIONS FOR CONSTRUCTIVE AND DESTRUCTIVE INTERFERENCE

10. Derive an expression for the intensity at any point on the observation screen in Young's double slit experiment. Hence write the conditions for constructive and destructive interference.

Expression for intensity at any point in interference pattern. Suppose the displacements of two light waves from two coherent sources S_1 and S_2 at point P on the observation screen at any time t are given by

$$y_1 = a_1 \sin \omega t$$

and

$$y_2 = a_2 \sin (\omega t + \phi)$$

where a_1 and a_2 are the amplitudes of the two waves, ϕ is the constant phase difference between the two waves. By the superposition principle, the resultant displacement at point P is

$$y = y_1 + y_2 = a_1 \sin \omega t + a_2 \sin (\omega t + \phi)$$

$$= a_1 \sin \omega t + a_2 \sin \omega t \cos \phi + a_2 \cos \omega t \sin \phi$$

or

$$y = (a_1 + a_2 \cos \phi) \sin \omega t + a_2 \sin \phi \cos \omega t$$

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$$\text{Put } a_1 + a_2 \cos \phi = A \cos \theta \quad \dots(1)$$

$$\text{and } a_2 \sin \phi = A \sin \theta \quad \dots(2)$$

$$\text{Then } y = A \cos \theta \sin \omega t + A \sin \theta \cos \omega t$$

$$\text{or } y = A \sin (\omega t + \theta)$$

Thus, the resultant wave is also a harmonic wave of amplitude A and it leads the first harmonic wave by phase angle θ . To determine A , squaring and adding equations (1) and (2), we get

$$A^2 \cos^2 \theta + A^2 \sin^2 \theta = (a_1 + a_2 \cos \phi)^2 + a_2^2 \sin^2 \phi$$

$$\text{or } A^2 = a_1^2 + a_2^2 (\cos^2 \phi + \sin^2 \phi) + 2a_1a_2 \cos \phi$$

$$\text{or } A^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos \phi \quad \dots(3)$$

But intensity of a wave $\propto (\text{amplitude})^2$

$$\text{We write } I = kA^2, \quad I_1 = ka_1^2 \quad \text{and} \quad I_2 = ka_2^2$$

where k is proportionality constant. The equation (3) can be written as

$$kA^2 = ka_1^2 + ka_2^2 + 2\sqrt{k}a_1 \sqrt{k}a_2 \cos \phi$$

$$\text{or } I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi \quad \dots(4)$$

This equation gives the total intensity at a point where the phase difference is ϕ . Here I_1 and I_2 are the intensities which the two individual sources produce on their own. The total intensity also contains a third term $2\sqrt{I_1 I_2} \cos \phi$. It is called *interference term*.

Constructive interference. The resultant intensity at the point P will be maximum when

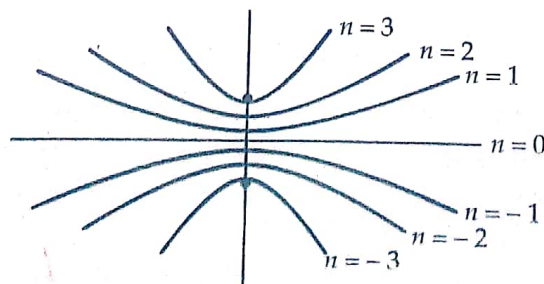
$$\cos \phi = 1 \quad \text{or} \quad \phi = 0, 2\pi, 4\pi, \dots$$

Since a phase difference of 2π corresponds to a path difference of λ , therefore, if p is the path difference between the two superposing waves, then

$$\frac{2\pi p}{\lambda} = 0, 2\pi, 4\pi, \dots$$

$$\text{or } p = 0, \lambda, 2\lambda, 3\lambda, \dots = n\lambda$$

Hence the resultant intensity at a point is maximum when the phase difference between the two superposing waves is an even multiple of π or path difference is an integral multiple of wavelength λ . This is the condition of constructive interference.



ig. 10.13 Locus of points for which $S_1P - S_2P$ is equal to $0, \pm\lambda, \pm2\lambda, \pm3\lambda$.

Destructive interference. The resultant intensity at the point P will be minimum when

$$\cos \phi = -1 \quad \text{or} \quad \phi = \pi, 3\pi, 5\pi, \dots$$

or
$$\frac{2\pi p}{\lambda} = \pi, 3\pi, 5\pi, \dots$$

or
$$p = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \dots = (2n-1)\frac{\lambda}{2}$$

Hence the resultant intensity at a point is minimum when the phase difference between the two superposing waves is an odd multiple of π or the path difference is an odd multiple of $\lambda/2$. This is the condition of destructive interference.

10.12 THEORY OF INTERFERENCE FRINGES : FRINGE WIDTH

13. Deduce an expression for fringe width in Young's double slit experiment. How can the wavelength of monochromatic light be found by this experiment?

Expression for fringe width in Young's double slit experiment. As shown in Fig. 10.14, suppose a narrow slit S is illuminated by monochromatic light of wavelength λ . S_1 and S_2 are two narrow slits at equal distance from S . Being derived from the same parent source S , the slits S_1 and S_2 act as two *coherent sources*,

separated by a small distance d . Interference fringes are obtained on a screen placed at distance D from the sources S_1 and S_2 .

Consider a point P on the screen at distance x from the centre O . The nature of the interference at the point P depends on path difference,

$$p = S_2P - S_1P$$

From right-angled ΔS_2BP and ΔS_1AP ,

$$S_2P^2 - S_1P^2 = [S_2B^2 + PB^2] - [S_1A^2 + PA^2]$$

$$= \left[D^2 + \left(x + \frac{d}{2} \right)^2 \right] - \left[D^2 + \left(x - \frac{d}{2} \right)^2 \right]$$

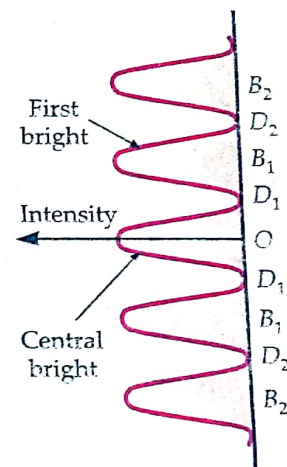
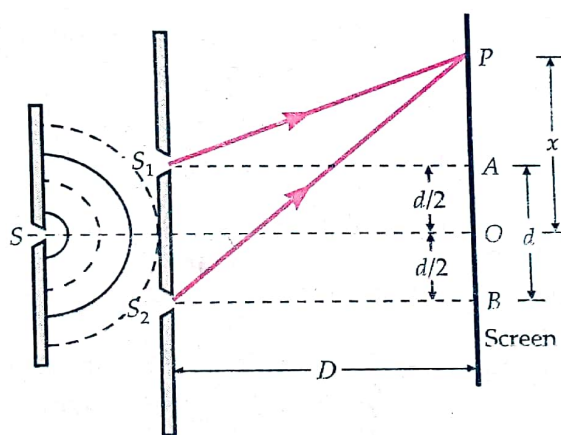


Fig. 10.14 Position of bright and dark fringes in Young's double slit experiment.

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$$\begin{aligned} \text{or } (S_2P - S_1P)(S_2P + S_1P) &= 2xd \\ \text{or } S_2P - S_1P &= \frac{2xd}{S_2P + S_1P} \end{aligned}$$

In practice, the point P lies very close to O , therefore $S_1P \approx S_2P \approx D$. Hence

$$p = S_2P - S_1P = \frac{2xd}{2D}$$

$$\text{or } p = \frac{xd}{D}$$

Positions of bright fringes. For constructive interference,

$$p = \frac{xd}{D} = n\lambda$$

$$\text{or } x = \frac{nD\lambda}{d} \quad \text{where } n = 0, 1, 2, 3, \dots$$

Clearly, the positions of various bright fringes are as follows :

For $n=0$, $x_0 = 0$ Central bright fringe

For $n=1$, $x_1 = \frac{D\lambda}{d}$ First bright fringe

For $n=2$, $x_2 = \frac{2D\lambda}{d}$ Second bright fringe

For $n=n$, $x_n = \frac{nD\lambda}{d}$ **n th bright fringe**

Positions of dark fringes. For destructive interference,

$$p = \frac{xd}{D} = (2n-1)\frac{\lambda}{2}$$

$$\text{or } x = (2n-1)\frac{D\lambda}{2d} \quad \text{where } n = 1, 2, 3, \dots$$

Clearly, the positions of various dark fringes are as follows :

For $n=1$, $x'_1 = \frac{1}{2} \frac{D\lambda}{d}$ First dark fringe

For $n=2$, $x'_2 = \frac{3}{2} \frac{D\lambda}{d}$ Second dark fringe

For $n=n$, $x'_n = (2n-1)\frac{D\lambda}{2d}$ **n th dark fringe**

Since the central point O is equidistant from S_1 and S_2 , the path difference p for it is zero. There will be a bright fringe at the centre O . But as we move from O upwards or downwards, alternate dark and bright fringes are formed.

Fringe width. It is the separation between two successive bright or dark fringes,

$$\begin{aligned} \text{Width of a dark fringe} &= \text{Separation between two consecutive bright fringes} \\ &= x_n - x_{n-1} = \frac{nD\lambda}{d} - \frac{(n-1)D\lambda}{d} = \frac{D\lambda}{d} \end{aligned}$$

Width of a bright fringe

= Separation between two consecutive dark fringes

$$= x'_n - x'_{n-1}$$

$$= (2n-1)\frac{D\lambda}{2d} - [2(n-1)-1]\frac{D\lambda}{2d} = \frac{D\lambda}{d}$$

Clearly, both the bright and dark fringes are of equal width.

Hence the expression for the fringe width in Young's double slit experiment can be written as

$$\beta = \frac{D\lambda}{d}$$

Slit width

As β is independent of n (the order of fringe), therefore, all the fringes are of equal width. In the case of light, λ is extremely small, D should be much larger than d , so that the fringe width β may be appreciable and hence observable.

Measurement of wavelength. Young's double slit experiment can be used to determine the wavelength of a monochromatic light. The interference pattern is obtained in the focal plane of a micrometer eyepiece and with its help fringe width β is measured. By measuring the distance d between the two coherent sources and their distance D from the eyepiece, the value of wavelength λ can be calculated as

$$\lambda = \frac{\beta d}{D}$$

For Your Knowledge

➤ In Young's double slit experiment, the width of the central bright fringe is equal to the distance between the first dark fringes on the two sides of the central bright fringe. So the **width of the central bright fringe** is given by

$$\beta_0 = 2x'_1 = 2 \times \frac{D\lambda}{2d} = \frac{D\lambda}{d}$$

➤ As all the bright and dark fringes are of the same width, the **angular width of a fringe** is given by

$$\theta = \frac{\beta}{D} = \frac{D\lambda/d}{D} = \frac{\lambda}{d}$$

➤ If Young's double slit apparatus is immersed in a liquid of refractive index μ , the wavelength of light decreases to $\lambda' (= \lambda/\mu)$ and so the fringe width reduces to

$$\beta' = \frac{D\lambda'}{d} = \frac{D\lambda}{\mu d} = \frac{\beta}{\mu}$$

Young's Double Slit Experiment

Formulae Used

1. For a bright fringe, path difference, $p = n\lambda$
2. For a dark fringe, $p = (2n - 1) \frac{\lambda}{2}$, $n = 1, 2, 3, \dots$
3. Distance of n th bright fringe from the centre of the screen,

$$x_n = n \frac{D\lambda}{d}, \quad n = 1, 2, 3, \dots$$

4. Distance of n th dark fringe from the centre of the screen,

$$x'_n = (2n - 1) \frac{D\lambda}{2d}$$

5. Fringe width, $\beta = \frac{D\lambda}{d}$

6. Wavelength of light used, $\lambda = \frac{\beta d}{D}$

7. Angular position of n th bright fringe,

$$\theta_n = \frac{x_n}{D} = \frac{n\lambda}{d}.$$

8. Angular position of n th dark fringe,

$$\theta'_n = \frac{x'_n}{D} = (2n - 1) \frac{\lambda}{2d}.$$

Units Used

Path difference p , distances x_n , x'_n , d and D ; wavelength λ and fringe width β are all in metre.

10.11 COHERENT & INCOHERENT SOURCES

11. What do you mean by coherent and incoherent sources of light? Why are coherent sources required to produce interference of light? Can two independent light sources be coherent?

Coherent and incoherent sources. Two sources of light which continuously emit light waves of same frequency (or wavelength) with a zero or constant phase difference between them, are called **coherent sources**.

Two sources of light which do not emit light waves with a constant phase difference are called **incoherent sources**.

Need of coherent sources for the production of interference pattern. When two monochromatic waves of intensity I_1 , I_2 and phase difference ϕ meet at a point, the resultant intensity is given by

$$I = I_1 + I_2 + 2 \sqrt{I_1 I_2} \cos \phi$$

The last term $2 \sqrt{I_1 I_2} \cos \phi$ is called **interference term**. There are two possibilities :

1. If $\cos \phi$ remains constant with time, the total intensity at any point will be constant. The intensity will be maximum $(\sqrt{I_1} + \sqrt{I_2})^2$ at points where $\cos \phi$ is $+1$ and minimum $(\sqrt{I_1} - \sqrt{I_2})^2$ at points where $\cos \phi = -1$. The sources in this case are coherent.

2. If $\cos \phi$ varies continuously with time assuming both positive and negative values, then the average value of $\cos \phi$ will be zero over time interval of measurement. Then interference term averages to zero. There will be same intensity, $I = I_1 + I_2$ at every point i.e., there will be general illumination on the observation screen. The two sources in the case are incoherent.

Hence to observe interference, we need to have two sources with the same frequency and with a stable phase difference. Such a pair of sources are called **coherent sources**.

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Two independent sources cannot be coherent. This is because of the following reasons :

1. Light is emitted by individual atoms and not by the bulk of matter acting as a whole.
2. Even a tiniest source consists of millions of atoms, and emission of light by them takes place independently.
3. Even an atom emits an unbroken wave of about 10^{-8} second due to its transition from a higher energy state to a lower energy state.

The millions of atoms of a source cannot emit waves in the same phase. The light emitted by the commonly used monochromatic source (a sodium lamp) remains coherent for about 10^{-8} s. After this time, the atoms responsible for emission of light get changed. The phase difference and hence the interference pattern changes 10^8 times in one second. Our eyes cannot see such rapid changes and a uniform illumination is seen on the screen. So two independent light sources cannot produce a sustained interference.

Two coherent sources can be obtained from a single parent source. Some of the methods of producing coherent sources are as follows :

1. In *Young's double slit experiment*, the two sources S_1 and S_2 get light from the same source S . Whatever phase changes occur in S_1 , the same phase changes occur in S_2 . The relative phase difference between S_1 and S_2 remains constant with time. So they act as coherent sources.
2. In *Fresnel's biprism method*, two coherent sources are obtained from the same parent source, by refraction.
3. In *Lloyd's mirror method*, a source and its reflected image act as two coherent sources.

12. State the conditions, which must be satisfied for two light sources to be coherent.

Conditions for obtaining two coherent sources of light :

1. The two sources of light must be obtained from a single source by some method. Then the relative phase difference between the two light waves from the sources will remain constant with time.
2. The two sources must give monochromatic light. Otherwise, different colours will produce different interference patterns and fringes of different colours will overlap.
3. The path difference between the waves arriving on the screen from the two sources must not be large. It should not exceed 30 cm. Then the phase difference produced due to path difference will not be constant. There will be general illumination on the screen.

10.13 CONDITIONS FOR SUSTAINED INTERFERENCE

14. What is a sustained interference pattern ? State the necessary conditions for obtaining a sustained interference of light.

Sustained interference pattern. In order to observe an interference pattern, it is necessary that the positions of maxima and minima do not keep on changing with time, otherwise the maxima and minima of intensity will mix up to produce uniform illumination. The interference pattern, in which the positions of maxima and minima of intensity on the observation screen do not change with time, is called a sustained or permanent interference pattern.

Conditions for sustained interference. The necessary conditions for obtaining a sustained and observable interference pattern of light are as follows:

1. The two sources should continuously emit waves of same frequency or wavelength.

2. The two sources of light should be *coherent*, i.e., they must vibrate either in the same phase or with a constant phase difference between them.
3. For a better contrast between maxima and minima of intensity, the amplitudes of the interfering waves should be equal.
4. The two sources should be narrow, otherwise interference will occur between waves of different parts of the same source and contrast will be poor.
5. The interfering waves must travel nearly along the same direction.
6. The sources should be *monochromatic*, otherwise fringes of different colours will overlap just to give a few observable fringes.
7. The interfering waves should be in the same state of polarisation.
8. To have sufficient fringe width, the distance between the two coherent sources should be small and the distance between the two sources and the screen should be large.

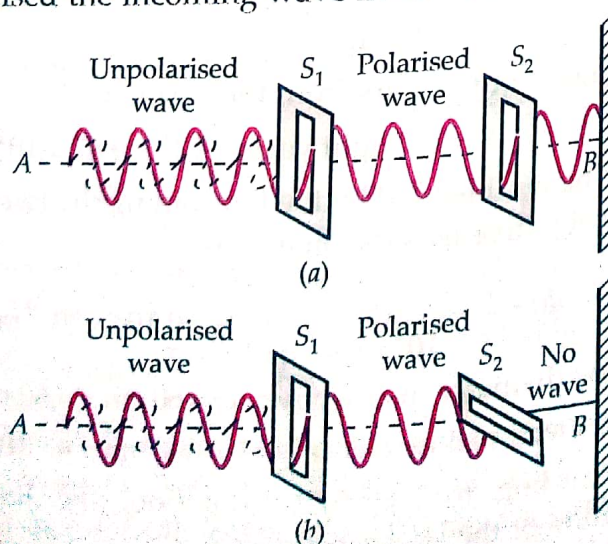
10.28 POLARISATION OF WAVES : INTRODUCTION

34. What are unpolarised and polarised waves? Explain polarisation, taking an example of mechanical waves. Can longitudinal waves be polarised?

Polarisation of waves. The waves are of two types: Transverse and Longitudinal. Both types of these waves undergo reflection, refraction, interference and diffraction. The difference is that only transverse waves can be polarised.

A transverse wave in which vibrations are present in all possible directions, in a plane perpendicular to the direction of propagation, is said to be **unpolarised**. If the vibrations of a wave are present in just one direction in a plane perpendicular to the direction of propagation, the wave is said to be **polarised** or **plane polarised**. The phenomenon of restricting the oscillations of a wave to just one direction in the transverse plane is called **polarisation of waves**.

Experimental demonstration with mechanical waves. Consider a long string AB passing through two rectangular slits S_1 and S_2 , as shown in Fig. 10.32. The end B of the string is tied to a hook in a wall and the free end A is jerked in all possible directions perpendicular to the length of the string so as to generate transverse waves in it. The portion AS_1 of the string has vibrations in all directions perpendicular to AB , so that the wave is unpolarised. The first slit S_1 will permit only those vibrations to pass through it which are parallel to the slit S_1 and will cut off all other vibrations. Thus the wave emerging from the slit S_1 is plane polarised. The slit S_1 is called the *polariser*. If the second slit S_2 , called the *analyser*, is held parallel to S_1 , the wave from S_1 will pass through S_2 unchanged. If S_2 is held perpendicular to S_1 , no vibrations will emerge from the slit S_2 . This indicates that the slit S_1 has polarised the incoming wave in the vertical plane.



Longitudinal waves cannot be polarised. This is because these waves are *symmetrical* about the direction of propagation. For example, if we pass a long spring through two slits and generate a longitudinal wave in it by alternately compressing and releasing its free end, it is seen that the compressions and rarefactions pass through the two slits, whatever is their relative orientation. This is so because the oscillations occur along the length of the spring, *i.e.*, along the direction of the wave propagation. On the other hand, the transverse waves can be polarised as they do not show any symmetry about the direction of wave propagation.

10.29 UNPOLARISED AND PLANE POLARISED LIGHT

35. What do you mean by unpolarised and plane polarised lights? Give their pictorial representations.

Unpolarised light. In ordinary light, electric field vector vibrates in all directions in a plane perpendicular to the direction of propagation. A light which has vibrations in all directions in a plane perpendicular to the direction of propagation is said to be **unpolarised light**. The light from the sun, a sodium lamp, an incandescent bulb or a candle is unpolarised. The electric field vector of such a light takes all possible directions in the transverse plane, rapidly and randomly, during the time of measurement, as shown in Fig. 10.33(a). The tip of the electric field vector traces an irregular curve, as shown in Fig. 10.33(b).

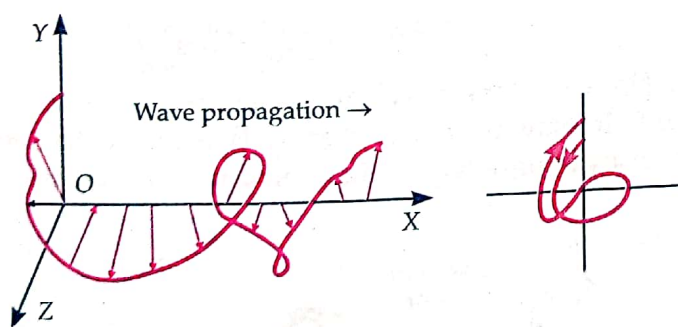


Fig. 10.33 (a) Random variation of the electric field vector of unpolarised light.

Fig. 10.33 (b) Irregular curve traced by the tip of this vector.

Figure 10.34(a) is the pictorial representation of unpolarised light propagating out of the plane of paper. It shows vibrations in all directions in the transverse plane. Figure 10.34(b) is also a pictorial representation of an unpolarised light. Here double arrows represent the vibrations in the plane of paper and small dots represent vibrations perpendicular to the plane of paper.

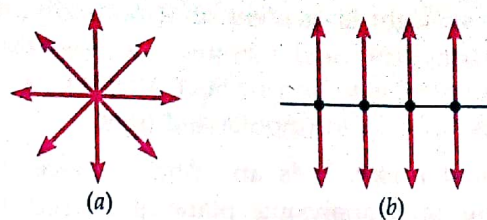


Fig. 10.34 Pictorial representations of unpolarised light.

Plane polarised or linearly polarised light. If the electric field vector of a light wave vibrates just in one direction perpendicular to the direction of wave propagation, then it is said to be **linearly polarised**. Since in a linearly polarised wave, the vibrations at all points, at all times, lie in the same plane, so it is also called a **plane polarised wave**. Fig. 10.35(a) shows the regular variation of the electric field vector of a linearly polarised light along Y-axis. Its tip vibrates back and forth along a straight line [Fig. 10.35(b)].

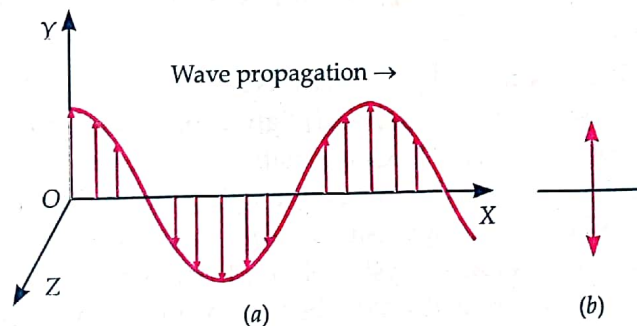


Fig. 10.35 (a) Regular variation of electric field vector of linearly polarised light. (b) The straight path traced by the tip of this vector.

Figs. 10.36 (a) and (b) show the pictorial representations for polarised light.

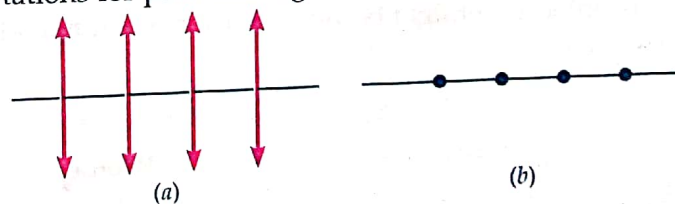


Fig. 10.36 Plane polarised light :
(a) Vibrations parallel to the plane of paper,
(b) Vibrations perpendicular to the plane of paper.

10.30 POLARISERS

36. What are polarisers? Name some commonly used polarisers.

Polarisers. A device that plane-polarises the unpolarised light passed through it is called a polariser.

Some commonly used polarisers are as follows :

1. **Tourmaline crystal.** The tourmaline crystal is so cut that its plane contains its optic axis. When

unpolarised light is incident on it normally, it allows only such electric field vibrations to pass through it which are parallel to its axis. Such a crystal can be used to polarise a beam of unpolarised light.

2. **Nicol prism.** It is an optical device used for producing and analysing plane polarised light. It consists of two pieces of calcite suitably cut and stuck together with Canada balsam.

3. **Polaroid.** A polaroid is a thin commercial sheet in the form of circular disc which makes use of the property of selective absorption to produce an intense beam of plane polarised light.

We shall study nicol prisms and polaroids in detail within next few sections.

10.31 EXPERIMENT TO DEMONSTRATE POLARISATION OF LIGHT

37. Describe an experiment to show that light waves are transverse in nature.

Polarisation of light waves. As shown in Fig. 10.37, when ordinary (unpolarised) light is passed through a tourmaline crystal P , its intensity is cut down to half. Rotate the crystal P about the incident beam. There is no effect on the intensity of the transmitted beam. Now, if a similar crystal A is placed with its axis parallel to that of P , all the light transmitted by P also passes through A . However, as A is rotated from this position in its own plane, the intensity of light transmitted by it goes on decreasing until it becomes zero, when the axes of the two crystals are perpendicular to each other. The two crystals are then said to be in the *crossed position*. In one full rotation of A , the intensity of light becomes twice maximum and twice zero.

Clearly, the light transmitted by crystal P is polarised as it contains vibrations parallel to the axis of P . When the axis of A is parallel to that of P , the polarised beam passes as such through A also. But when the axis of A is perpendicular to that of P , the vibrations of the waves (being perpendicular to the axis of A) are not transmitted by A . The first crystal P which polarises the light is called *polariser* and the second crystal is called the *analyser*, because it analyses whether the light is polarised or not.

Since the polarisation of light can neither be explained on the basis of corpuscular theory of light nor by assuming that light propagates as longitudinal waves, therefore, the above experiment proves that light propagates in the form of transverse waves.

10.32 LAW OF MALUS

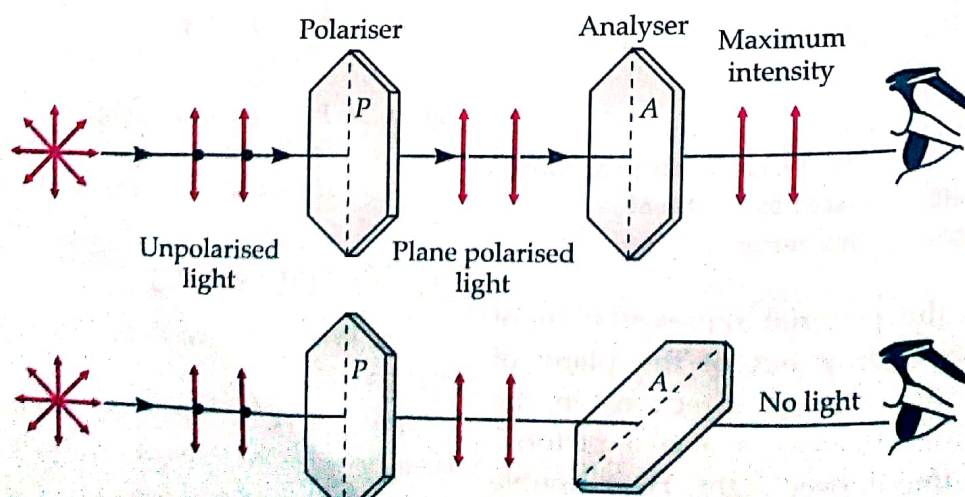
38. State and explain the law of Malus. Sketch a graph showing the dependence of intensity of transmitted light on the angle between polariser and analyser.

Law of Malus. When a plane polarised light is seen through an analyser, the intensity of transmitted light varies as the analyser is rotated in its own plane about the incident direction. In 1809, E.N. Malus discovered that when a beam of completely plane polarised light is passed through analyser, the intensity ' I ' of transmitted light varies directly as the square of the cosine of the angle ' θ ' between the transmission directions of polariser and analyser. This statement is known as the *law of Malus*.

Mathematically,

$$I \propto \cos^2 \theta \quad \text{or} \quad I = I_0 \cos^2 \theta$$

Here I_0 is the maximum intensity of transmitted light. It may be noted that I_0 is equal to half the intensity of unpolarised light incident on the polariser.



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Explanation of the law. As shown in Fig. 10.38, suppose that the planes of polariser and analyser are inclined to each other at an angle θ . Let I_0 be the intensity of the plane polarised light transmitted by the polariser.

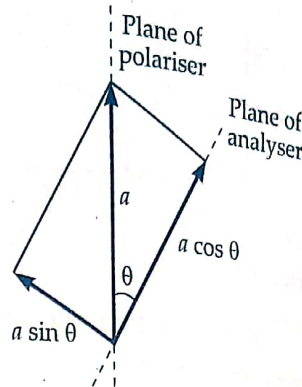


Fig. 10.38 Law of Malus.

The amplitude a of the light incident on the analyser has two rectangular components :

1. $a \cos \theta$, parallel to the plane of transmission of the analyser, and
2. $a \sin \theta$, perpendicular to the plane of transmission of the analyser.

So only the component $a \cos \theta$ is transmitted by the analyser. The intensity of light transmitted by the analyser is

$$I = k (a \cos \theta)^2 = k a^2 \cos^2 \theta$$

or $I = I_0 \cos^2 \theta$

where $I_0 = k a^2$, is the maximum intensity of light transmitted by the analyser (when $\theta = 0^\circ$). The above equation is the **law of Malus**.

Special Cases

1. When $\theta = 0^\circ$ or 180° , $\cos \theta = \pm 1$, so that $I = I_0$

So when the transmission directions of polariser and analyser are parallel or antiparallel to each other, the maximum intensity of plane polarised light is transmitted by the analyser and is equal to the intensity emerging from the polariser.

2. When $\theta = 90^\circ$, $\cos \theta = 0$, so that $I = 0$

So when the transmission directions of polariser and analyser are perpendicular to each other, the intensity of light transmitted through the analyser is zero.

3. When a beam of unpolarised light is incident on the polariser,

$$I = I_0 \overline{\cos^2 \theta} = I_0 \times \frac{1}{2} (1 + \cos 2\theta)$$

$$= \frac{1}{2} I_0 (1 + \overline{\cos 2\theta}) = \frac{1}{2} I_0 (1 + 0) = \frac{1}{2} I_0$$

between the

10.48

10.33 PLANES OF POLARISATION AND VIBRATION

39. With the help of a diagram, explain the plane of vibration and plane of polarisation.

Plane of vibration and plane of polarisation. When ordinary light is passed through a tourmaline crystal, the light is plane polarised and the vibrations of the electric field vector take place just in one direction perpendicular to the direction of propagation of light.

The plane containing the direction of vibration and the direction of wave propagation is called the **plane of vibration**.

As shown in Fig. 10.42, if light wave is propagating along X-axis and the electric field vector vibrates parallel to Y-axis, then ABCD or XY-plane is the plane of vibration.

The plane passing through the direction of wave propagation and perpendicular to the plane of vibration is called the **plane of polarisation**. No vibrations occur in the plane of polarisation.

In Fig. 10.42, EFGH or XZ-plane is the plane of polarisation.

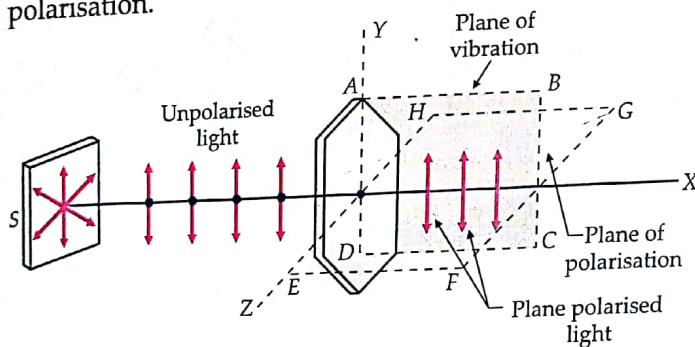


Fig. 10.42 Plane of vibration and plane of polarisation.

10.34 CIRCULARLY AND ELLIPTICALLY POLARISED LIGHTS

40. What do you understand by circularly and elliptically polarised lights?

Circularly polarised light. If the tip of the electric field vector of a light wave traces a circle, the light is said to be circularly polarised. It can be regarded as the combination of two plane polarised vibrations of equal amplitudes in two mutually perpendicular directions with a phase difference of $\pi/2$.

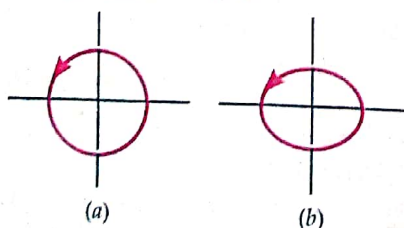


Fig. 10.43 Paths traced by the electric field vectors of (a) circularly polarised and (b) elliptically polarised lights.

Elliptically Polarised light. If the tip of the electric field vector of a light wave traces an ellipse, the light is said to be elliptically polarised. It can be regarded as the combination of two plane polarised vibrations of unequal amplitudes in the mutually perpendicular directions with a phase difference of $\pi/2$.

10.35 METHODS OF PRODUCING PLANE POLARISED LIGHT

41. Name the various methods by which ordinary light can be plane polarised.

Methods of producing plane polarised light. Ordinary light can be polarised by using any of the following phenomena :

1. Reflection
2. Scattering
3. Double refraction
4. Selective absorption.

10.36 POLARISATION BY REFLECTION : BREWSTER LAW

42. Explain polarisation by reflection. State and prove Brewster law of polarisation.

Polarisation by reflection. In 1808, the French physicist *Malus* discovered that when ordinary light is incident on the surface of a transparent medium, the reflected light is partially plane polarised. The extent of polarisation depends on the angle of incidence. For a particular angle of incidence, the reflected light is found to be completely polarised with its vibrations perpendicular to the plane of incidence.

The angle of incidence at which a beam of unpolarised light falling on a transparent surface is reflected as a beam of completely plane polarised light is called **polarising** or **Brewster angle**. It is denoted by i_p .

The British physicist *David Brewster* found that at the polarising angle, the reflected and transmitted rays are perpendicular to each other, as shown in Fig. 10.44. Suppose i_p is the polarising angle of incidence and r_p the corresponding angle of refraction. Then

$$i_p + r_p = 90^\circ \quad \text{or} \quad r_p = 90^\circ - i_p$$

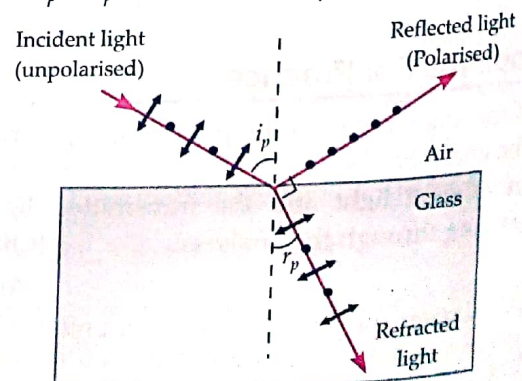


Fig. 10.44 Polarisation of light reflected from a transparent medium at the Brewster angle.

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From Snell's law, the refractive index of the transparent medium is

$$\mu = \frac{\sin i_p}{\sin r_p} = \frac{\sin i_p}{\sin(90^\circ - i_p)} = \frac{\sin i_p}{\cos i_p} \quad \text{or} \quad \mu = \tan i_p$$

This relation is known as **Brewster law**. The law states that the tangent of the polarising angle of incidence of a transparent medium is equal to its refractive index.

The value of Brewster angle depends on the nature of the transparent refracting medium and the wavelength of light used.

Explanation. As shown in Fig. 10.44, the incident unpolarised light has both types of vibrations, one perpendicular (dots) and other parallel to the plane of incidence. At the polarising angle of incidence (i_p), the reflected and refracted rays are perpendicular to each other. The electrons oscillating in the transparent medium produce the reflected wave. These vibrations move in two directions transverse to the refracted wave. As the arrows are parallel to the direction of reflected wave, they cannot send energy along the direction of reflected light. Hence the reflected light consists of vibrations perpendicular to the plane of incidence (dots) only i.e., the reflected light is plane polarised.

Examples based on

Brewster Law

Formulae Used

1. Brewster law, $\mu = \tan i_p$
2. $i_p + r_p = 90^\circ$
3. ${}^2\mu_3 = \frac{{}^1\mu_2}{{}^1\mu_3}$

Units Used

Angle i_p and r_p are in degrees, refractive index μ has no units.

... light is incident on a plane ... so that the

10.41 USES OF POLAROIDS

48. Explain some of the important uses of polaroids.

Uses of polaroids. Polaroids have several uses in daily life :

1. *In sunglasses and camera filters.* Sunglasses and camera filters are made of polarising sheets to reduce the glare of light produced by reflection from shiny surface such as water surface.

2. *In wind screens.* The wind screens and car head lights of motor cars are fitted with polaroid films with their axes inclined at 45° to the horizontal. When two cars approach each other from opposite directions, the transmission planes of their wind screens will be perpendicular to each other, so the glare of their head lights is completely eliminated. Each driver sees the road by the light sent by his own car.

3. *In window panes of aeroplanes.* One of the polaroids is fixed while the other can be rotated to control the amount of light coming in.

4. *In photoelasticity.* Glass and some plastic materials exhibit double refraction only when stressed. If polarised light is passed through them and then

analysed, the bright coloured lines indicate the existence of strains. In engineering work, plastic models of structures are constructed and weaknesses are examined in this way.

5. *In three-D movies.* Three-D motion pictures are projected on screen by two projectors, each forming a slightly different image. One image is for one eye and other for the second eye so that the brain interprets this difference as depth or third dimension. Lights from each projector are plane polarised, but in mutually perpendicular directions. The two 3-D glasses are really polarising glasses with their directions of polarisation perpendicular to each other. So one eye sees one image and other sees a slightly different image.

6. *In liquid crystal displays (LCDs).* An important application of polarisation is in liquid crystal displays or LCDs, used in many watches, calculators and portable computers (lap tops). Liquid crystals have long molecules whose directions can be controlled by applying electric fields. This fact is used in rotating the plane of polarisation of light produced by a polariser so that its polarisation is perpendicular to the axis of an analyser which cuts it out. These dark regions can be controlled with applied voltages and used to form letters and numbers.

10.40 POLAROIDS

47. What are polaroids ? How do they plane polarise an unpolarised beam of light ?

Polaroids. Polaroids are thin commercial sheets which make use of the property of selective absorption to produce an intense beam of plane polarised light.

In 1932, an American scientist *Edwin Land* developed a polariser in the form of large sheets. When a paste of quinine iodosulphate made in nitrocellulose is squeezed out through a fine slit, the needle-shaped crystals of quinine iodosulphate align themselves parallel to their optic axis. These crystals are highly dichroic. They absorb one of the doubly refracted beams completely. The thin polarising sheet so obtained is enclosed between two thin glass plates for mechanical support and we get a polaroid. Each polaroid has a characteristic

